MODULE 17 · DOMAIN 2 - RELIABILITY AND BUSINESS CONTINUITY

Disaster Recovery

RPO, RTO, the four strategies AWS names explicitly - and Amazon Application Recovery Controller.

DOMAIN 2 of 5 EXAM WEIGHT 22% MODULE 17 of 35

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DR is a business decision expressed as two numbers. **RPO** is how much data you can lose. **RTO** is how long you can be down. The strategy follows.

RPO and RTO, precisely

- **RPO looks backwards.** Nightly backups mean an RPO of up to 24 hours, because a failure at 5pm loses the whole day.
- **RTO looks forwards.** It is the time from the failure to the service being usable, not to the restore finishing.
- **They are set by the business**, per workload. A payments system and an internal wiki should not share a target.
- A target you have never **measured** is an aspiration. Module sixteen's restore testing is how you turn one into a number.

> ON THE EXAM
>
> RPO is decided by how often you replicate. RTO is decided by how much has to happen after the failure. They are tuned by different levers.

The four strategies

STRATEGY	TYPICAL RPO AND RTO, AND WHAT RUNS
Backup and restore	RPO hours, RTO hours to days. Nothing runs in the DR Region
Pilot light	RPO minutes, RTO tens of minutes. Data replicated; compute switched off
Warm standby	RPO seconds, RTO minutes. A scaled-down but working copy runs
Active/active	RPO and RTO near zero. Full capacity serving in both Regions

ON THE EXAM

Cost rises down the table and recovery time falls. There is no strategy that is both cheapest and fastest, and questions test whether you accept that.

Backup and restore

- Copy backups to the second Region. Nothing else exists there until you need it.
- Cheapest by a wide margin, and the correct answer for workloads that can be down for a day.
- **Recovery includes building the infrastructure**, which is why infrastructure as code turns days into hours here.
- Watch **service quotas** in the recovery Region. Quotas are per Region, and a fresh Region has default limits.

> ON THE EXAM
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> The RTO here is dominated by everything that is not the data: VPCs, load balancers, instances, DNS. Templates are the difference between hours and days.

Pilot light

- **The data layer is live and replicating.** The compute layer exists as configuration but is switched off.
- Typically: a cross-Region read replica or global table running, AMIs and launch templates ready, Auto Scaling groups at zero.
- Recovery means **promoting the database and scaling compute up from zero**.
- You pay for storage and replication continuously, and for compute only during a disaster or a test.

> ON THE EXAM
>
> The name is literal. A pilot light is the small permanent flame you use to light the main burner quickly. The data is the flame.

Warm standby

- A **complete, working copy** of the environment runs in the second Region, at reduced capacity.
- Because it is running, it can **serve a little real traffic**, which proves continuously that it works.
- Recovery means **scaling it up** and shifting traffic. There is nothing to build and nothing to promote from cold.
- More expensive than pilot light, because you are paying for running compute you are not fully using.

> ON THE EXAM
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> The decisive advantage is not speed. It is that a warm standby is tested by simply existing, whereas a pilot light is tested only when you deliberately test it.

Active/active

- Both Regions serve production traffic **all the time**. A failure removes capacity rather than causing an outage.
- Requires **multi-Region data**: DynamoDB global tables, Aurora Global Database, or S3 replication.
- The hard part is **write conflicts and consistency**, not infrastructure. Two Regions accepting writes is an application design problem.
- Most expensive, and it constrains how you build the application from the beginning.

> ON THE EXAM
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> This is the only strategy where a Region failure is a capacity event rather than a recovery event. That is what you are paying for.

Pilot light or warm standby?

PILOT LIGHT

Data warm, compute cold

- Compute is off; nothing serves traffic
- Recovery: promote the database, scale from zero
- Cheaper: you pay for data, not compute
- Untested unless you deliberately test it

WARM STANDBY

Everything on, but small

- A scaled-down environment is running and serving
- Recovery: scale up and shift traffic
- More expensive: running compute all the time
- Continuously proven, because it is in use

THE TIEBREAKER

Ask whether anything is running and serving in the second Region. If yes, warm standby. If only the data is live, pilot light.

Getting the data there

SERVICE	CROSS-REGION MECHANISM
Aurora	Global Database. RPO about 1 second, managed failover in about a minute
RDS	Cross-Region read replica, or cross-Region automated backups
DynamoDB	Global tables. Multi-active, last writer wins
S3	Cross-Region Replication. Replication Time Control gives a 15-minute SLA
EBS and AMIs	Snapshot and AMI copy, or AWS Backup cross-Region copy
ECR	Cross-Region and cross-account replication rules

ON THE EXAM

Replication frequency sets your RPO. Everything in this table except backup copying gives you an RPO measured in seconds or minutes.

The half that is not data

- **AMIs and container images** must exist in the recovery Region. Copy them on a schedule, not during an incident.
- **Infrastructure as code**, deployable into the second Region. StackSets or a pipeline, not a template nobody has run.
- **Certificates**: ACM certificates are **Regional**. A certificate for the primary Region does not exist in the secondary.
- **Secrets and parameters**: Secrets Manager replicates; Parameter Store does not, so plan for it.

> ON THE EXAM
>
> Most failed DR tests fail here, not on the data. The database came up fine and there was no AMI, no certificate, and no route to anything.

Service quotas: the classic failure

- **Quotas are per account per Region.** Your recovery Region has default limits, not the limits you raised over three years.
- Common ones: EC2 vCPU limits by instance family, Elastic IPs, VPCs, security group rules, Lambda concurrency.
- **Raise them in advance.** Quota increases are handled by AWS and can take time you will not have.
- This is exactly what ARC readiness check was built to monitor, and what Region switch plan evaluation now covers.

> ON THE EXAM
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> A pilot light that cannot scale past 20 vCPUs is not a recovery plan. The quota is invisible until the moment it matters most.

Amazon Application Recovery Controller

- Formerly **Route 53 Application Recovery Controller**. Two domains: **Multi-AZ recovery** and **Multi-Region recovery**.
- **Multi-AZ**: zonal shift and zonal autoshift, for a single impaired Availability Zone.
- **Multi-Region**: routing controls, Region switch, and readiness check.
- Its point is a **data plane that keeps working while a Region is impaired**, which ordinary control planes may not.

> ON THE EXAM
>
> Most resilience events are zonal, not Regional. The Multi-AZ half of ARC is the part most workloads actually need.

Zonal shift and zonal autoshift

- **Zonal shift** is manual: you move traffic for a supported resource away from one Availability Zone, temporarily.
- **Zonal autoshift** is automatic: AWS shifts traffic away when its **internal telemetry** indicates an impairment.
- AWS's telemetry draws on the network, EC2 and Elastic Load Balancing, so it sees things your own metrics may not.
- **Both are temporary.** AWS ends an autoshift once the indicators clear. Neither is a permanent configuration change.

> ON THE EXAM
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> Requires capacity in the remaining zones. Autoshift assumes you are statically stable enough to lose a zone, which is module 14's point again.

Routing controls and safety rules

- A **routing control** is a simple on/off switch that changes the health of a Route 53 record. Flipping it moves traffic.
- It is hosted on a **cluster of five redundant endpoints in five Regions**, so you can operate it during a Regional event.
- **Safety rules** prevent dangerous combinations: an assertion rule can stop you turning off both Regions at once.
- A cluster costs by the hour and there is a **maximum of two per account**, so this is a deliberate, funded design choice.

> ON THE EXAM
>
> The point is not that a switch flips DNS. It is that the switch itself is reachable when the thing you are failing away from is broken.

Region switch, and one closure

- **Region switch** orchestrates a full multi-Region recovery: a plan, executed across multiple accounts, centrally and observably.
- It includes **plan evaluation**, which continually checks that the plan is still executable.
- **ARC readiness check closed to new customers on 30 April 2026.** Existing customers continue.
- **ARC itself and Region switch are fully supported.** Only readiness check is affected, and Region switch is the recommended replacement.

> ON THE EXAM
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> The third "still examinable, no longer buildable" item in this course, after CloudTrail Lake and Systems Manager Incident Manager. Learn it, do not design on it.

Failback, and testing

- **Failback is harder than failover.** The primary Region now has stale data, and reversing replication is rarely symmetrical.
- Decide in advance whether you fail back at all, or simply run in the recovery Region until a planned move.
- **Game days** exercise the whole plan, including the human parts: who decides, who has access, who is awake.
- **AWS Fault Injection Service** injects controlled failures, so you test the response rather than assuming it.

> ON THE EXAM
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> A plan that has only ever been read is not a plan. Most of what goes wrong in a real failover is organisational, not technical.

A company must recover a payments application in a second Region. The RPO is under 5 seconds and the RTO under 10 minutes. Cost is a secondary consideration. Writes must continue to be served from a single Region to avoid conflict handling.

The application runs on EC2 behind an ALB with an Aurora PostgreSQL cluster.

Which strategy should the company implement?

A. Backup and restore, with AWS Backup cross-Region copy

B. Pilot light, with an Aurora Global Database and Auto Scaling groups at zero

C. Warm standby, with an Aurora Global Database and a scaled-down running environment

D. Active/active, with writes accepted in both Regions

◆ PAUSE AND DECIDE

Which strategy should the company implement?

A Backup and restore

RPO and RTO measured in hours. Both requirements are missed by two orders of magnitude.

B Pilot light

The genuinely close alternative. The Global Database meets the RPO, but scaling compute from zero and warming caches makes 10 minutes tight.

C Warm standby

Global Database gives roughly a one-second RPO; a running scaled-down environment needs only scaling up, which fits comfortably inside 10 minutes.

D Active/active

Explicitly excluded. The requirement says writes must come from one Region to avoid conflict handling, which is what active/active demands.

THE SIGNAL

Three clauses each eliminate something: the RPO kills A, the RTO makes B risky, and the single-writer constraint kills D.

A team tests its pilot light DR plan for the first time. The Aurora Global Database promotes successfully in 45 seconds. The Auto Scaling group then fails to launch beyond 8 instances, reporting a vCPU limit error, when 60 are required.

The primary Region has had its limits raised repeatedly over four years.

What is the cause and the fix?

A The launch template references an AMI that does not exist in the DR Region

B Service quotas are per Region. The DR Region still has default vCPU limits, and increases must be requested in advance

C The Auto Scaling group's maximum capacity is set too low

D Aurora Global Database promotion blocks compute scaling until replication completes

◆ PAUSE AND DECIDE

What is the cause and the fix?

A **Missing AMI**

A real and common DR failure, and it would fail every launch with an AMI error. Eight launched successfully, so the AMI exists.

B **Per-Region quotas at defaults**

The error names a vCPU limit, and quotas are per account per Region. Four years of increases in the primary Region mean nothing in the secondary.

C **Maximum capacity too low**

Would fail silently at the ceiling, not produce a vCPU limit error. The error text names the actual cause.

D **Promotion blocks scaling**

Invented behaviour. The scenario also states promotion completed successfully in 45 seconds.

THE SIGNAL

Eight instances launching is the clue. A missing AMI or a broken template fails at zero; a quota fails partway through.

During a Regional event, an operations team needs to shift traffic away from the impaired Region. Their failover mechanism updates Route 53 records using the AWS Management Console and the Route 53 API from the affected Region's console session.

The team wants a mechanism they can rely on to work while a Region is degraded, and wants to prevent anyone accidentally disabling both Regions.

What should the team implement?

A. ARC routing controls on a cluster, with a safety rule preventing both Regions being turned off

B. Route 53 failover routing with health checks on both Regions

C. ARC zonal autoshift on the load balancers in both Regions

D. A Lambda function in the secondary Region that rewrites Route 53 records

◆ PAUSE AND DECIDE

What should the team implement?

A Routing controls with a safety rule

The cluster spans five Regions, so the switch is operable during a Regional event, and safety rules block the both-off state.

B Failover routing with health checks

Useful and automatic, and it gives no manual control and no safety rules. The team asked for a mechanism they operate deliberately.

C Zonal autoshift

Addresses a single impaired Availability Zone within a Region. This is a Regional event, which is the other half of ARC.

D Lambda rewriting records

Builds the switch by hand, with no safety rules, and depends on Lambda and the Route 53 control plane behaving during the event.

THE SIGNAL

Two requirements: operable during a Regional impairment, and impossible to turn everything off. Only routing controls answer both.

> An internal HR reporting system has an agreed RTO of 24 hours and an RPO of 12 hours. It runs on two EC2 instances and an RDS for MySQL database. The finance director has asked why the team is proposing to spend money on a second Region at all.
>
> The team must protect against the loss of the primary Region.

What should the team propose?

A Warm standby, because it is the safest option

B Backup and restore, using AWS Backup cross-Region copy and CloudFormation templates for the infrastructure

C Pilot light, with a cross-Region read replica running continuously

D No DR at all, since the system is internal

◆ PAUSE AND DECIDE

What should the team propose?

A **Warm standby**

Massively over-engineered for a 24-hour RTO, and it spends continuously on running compute nobody needs.

B **Backup and restore with templates**

Matches the agreed targets at the lowest cost. Templates are what keep the rebuild inside 24 hours rather than stretching to days.

C **Pilot light**

Better than needed. A continuously running cross-Region replica buys an RPO of seconds when 12 hours was agreed.

D **No DR**

The requirement explicitly says the loss of the Region must be survivable. Internal does not mean expendable.

THE SIGNAL

Over-engineering is a wrong answer, not a safe one. The targets are agreed, and exceeding them costs money the business chose not to spend.

> A team designs a new multi-Region recovery plan in 2026 and wants continuous confirmation that the secondary Region remains capable of taking over: correct quotas, sufficient capacity, and matching routing configuration.
>
> An architect recommends ARC readiness checks, which they used at a previous employer.

What should the team do?

A Implement ARC readiness checks as recommended

B Onboard to ARC Region switch and use its plan evaluation capability

C Use Route 53 health checks against the secondary Region's endpoint

D Use AWS Config conformance packs in the secondary Region

◆ PAUSE AND DECIDE

What should the team do?

A ARC readiness checks

Closed to new customers on 30 April 2026. A team designing a new plan cannot onboard, whatever the architect used previously.

B Region switch plan evaluation

AWS's own recommended replacement. Plan evaluation continually monitors that the recovery plan remains executable.

C Route 53 health checks

Confirms an endpoint responds. It says nothing about quotas, capacity or configuration parity.

D Config conformance packs

Checks resource configuration against rules, which helps. It does not evaluate quotas, capacity or readiness to fail over.

THE SIGNAL

The same shape as the Incident Manager scenario in module 7: an experienced recommendation that is no longer available to a new customer.

The five things to carry forward

- **RPO is set by replication frequency; RTO by what must happen after the failure.** Different levers.
- **Is anything running and serving in the second Region?** Yes is warm standby, no is pilot light.
- **Over-engineering is a wrong answer.** Meet the agreed targets, do not exceed them expensively.
- **Quotas, AMIs and certificates are per Region.** That is what fails a DR test, not the data.
- **Zonal shift for one AZ, routing controls and Region switch for a Region.** Readiness check is closed to new customers.

> ON THE EXAM
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> That completes domain two. Next: CloudFormation, and the start of domain three.